Dplyr Example for R Workshop 7/10/26

Dplyr

Dplyr is a useful package within the tidyverse suite of packages for R Studio. Dplyr is particularly helpful with data management. Below is an example of common uses for the functions within Dplyr.

1. Install/Load the Dplyr Package

```
#remove the hashtag below before running
#install.packages("dplyr")
library(dplyr)
library(readr)
library(readxl)
```

2. Load the Dataset

```
#defining a variable called "mydata" that contains my csv datasheet
#Copy the pathname of your data file and paste in the parentheses embedded
#within single apostrophes
mydata <- read_csv('/Users/colinscholl/Downloads/AXCPT_Merge001-014.csv')
#read_excel() can be used for opening data from an excel sheet
```

3. Perform Dplyr Functions

The pipe operator is %>%. %>% means that any manipulation by Dplyr will be applied to the the data variable on the left. Below are some examples of common functions in Dplyr.

MUTATE

Mutate is a function that adds computes and appends new column variables to a dataset.

```
#adding a column "checkifprac" to see if the trial is a practice or
#experimental trial, 0 is a practice trial and 1 is an experimental trial
mydata_mutate <- mydata %>% mutate(checkifprac = ifelse(Procedure == "ExpProc", 1, 0))
```

FILTER

```
#filtering to only include experimental trials
mydata_filter <- mydata %>% filter(Procedure == "ExpProc")
```

SELECT

```
#selecting only the specific columns "Category" and "Stimulus1.ACC"
mydata_select <- mydata %>% select(Category, Stimulus1.ACC)
```

GROUP_BY

```
#Grouping the data by "Category"
mydata_groupby <- mydata %>% group_by(Category)
```

SUMMARISE

```
#Summarise creates a unique table with user-created variables
#around the data. This code creates a 1x1 tibble with the mean
#accuracy of all trials.
mydata_summarise <- mydata %>% summarise(avg_accuracy = mean(Stimulus1.ACC, na.rm = TRUE))
```

ARRANGE

```
#sorting the data according to descending trial order.
#To make ascending, remove desc().
mydata_arrange <- mydata %>% arrange(desc(Trial))
```

EXAMPLE

```r
#This code finds the accuracy of the experimental trials
#for each of the four trial conditions
mydata_example <- mydata %>% filter(Procedure == "ExpProc") %>%
            select(Category, Stimulus1.ACC) %>% group_by(Category) %>%
            summarise(avg_accuracy = mean(Stimulus1.ACC))
mydata_example
```

```
# A tibble: 4 x 2
  Category avg_accuracy
  <chr>           <dbl>
1 AX              0.905
2 AY              0.471
3 BX              0.894
4 BY              0.861
```